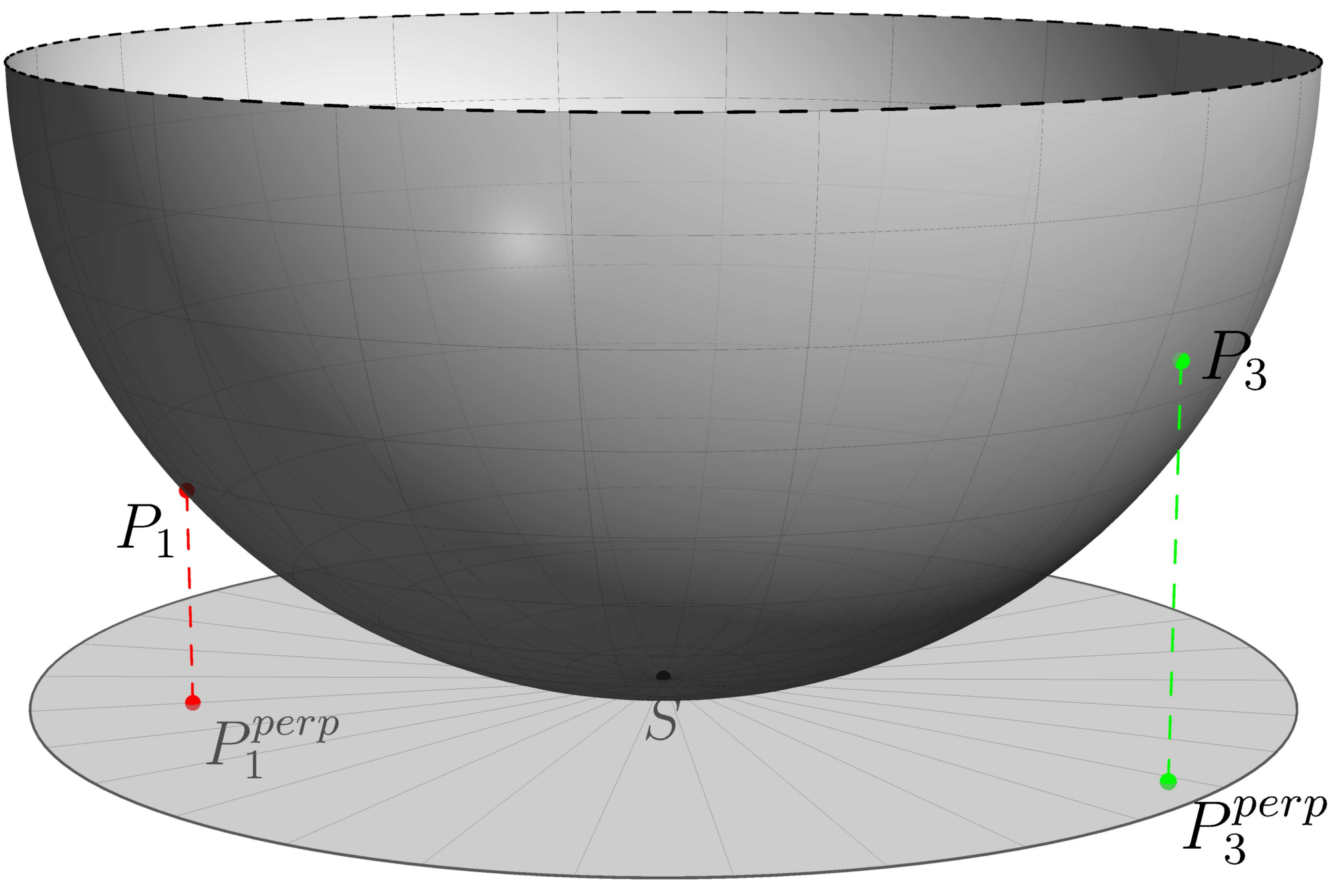

P_1
P_1^{perp}
S
P_3
P_3^{perp}